Lab experiment

8--queen problem

Coding:

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# 8-Queen Problem using Backtracking

N = 8
board = [-1] * N

# Check whether placing a queen is safe
def is_safe(row, col):
    for i in range(row):
        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True

# Solve the 8-Queen Problem
def solve(row):
    if row == N:
        print_solution()
        return True

    for col in range(N):
        if is_safe(row, col):
            board[row] = col

            if solve(row + 1):
                return True

            board[row] = -1

    return False

# Display the solution
def print_solution():
    for row in range(N):
        for col in range(N):
            if board[row] == col:
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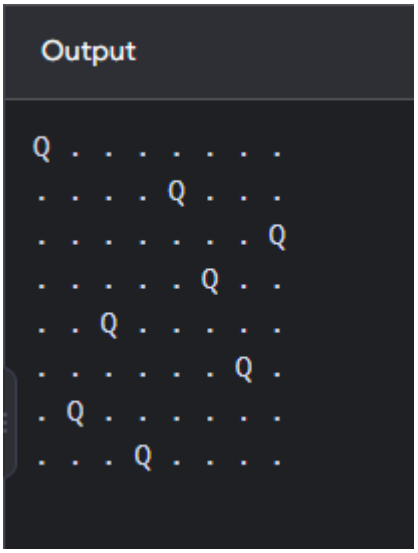
            print("Q", end=" ")
        else:
            print(".", end=" ")
    print()

# Main program
if not solve(0):
    print("No solution exists")
```

Output: